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Math 3108 - Section 103 (Murphy) - Fall 2019 - Exam 1

Name.

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Instructions.

- No outside notes or resources.
- Silence your phones completely (including vibrations).
- Be mindful of your classmates.
- No calculators. You need to show your work.
- There are 6 problems for a total of 100 points.
- You have 1 hour to complete the exam.

$$x_1 - 2x_2 + 6x_3 = 5 \Rightarrow x_1 = 2x_2 - 6x_3 + 5$$

$$x_1 - 2x_2 = 3 \Rightarrow x_1 = 2x_2 + 3$$

$$\vec{x} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -6 \\ 0 \\ 1 \end{bmatrix}$$

1. (20 points) Find the general solution of the following linear system:

$$x_1 - 7x_2 + 6x_4 = 5,$$

$$x_3 - 2x_4 = -3,$$

$$-x_1 + 7x_2 - 4x_3 + 2x_4 = 7.$$

Write the solution in parametric vector form.

$$\begin{bmatrix} 1 & -7 & 0 & 6 & | & 5 \\ 0 & 0 & 1 & -2 & | & -3 \\ -1 & 7 & -4 & 2 & | & 7 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -7 & 0 & 6 & | & 5 \\ 0 & 0 & 1 & -2 & | & -3 \\ 0 & 0 & -4 & 8 & | & 12 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -7 & 0 & 6 & | & 5 \\ 0 & 0 & 1 & -2 & | & -3 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$x_4 = \text{free}$$

$$x_2 = \text{free}$$

$$\text{if } x_4 = 0$$

$$x_3 = 0$$

$$x_1 = 5$$

$$x_3 = -3$$

$$\begin{bmatrix} x_1 = 5 \\ x_2 = 0 \\ x_3 = -3 \\ x_4 = 0 \end{bmatrix}$$

$$x_1 - 7x_2 + 6x_4 = 5 \rightarrow x_1 = 7x_2 - 6x_4 + 5$$

$$x_3 - 2x_4 = -3 \rightarrow x_3 = 2x_4 - 3$$

$$\vec{x} = \begin{bmatrix} 5 \\ 0 \\ -3 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 7 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -6 \\ 0 \\ 2 \\ 1 \end{bmatrix}$$

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2. (20 points) Define the vectors v_1, v_2, v_3 , and b by

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 3 \\ 4 \\ 2 \end{bmatrix}, \quad b = \begin{bmatrix} -2 \\ -1 \\ -3 \end{bmatrix}.$$

(i) Does b belong to $\text{Span}\{v_1, v_2, v_3\}$? Justify your answer.

(ii) Is $\{v_1, v_2, v_3\}$ a linearly independent set? Justify your answer.

$$(i) \left[\begin{array}{ccc|c} 1 & 1 & 3 & -2 \\ 2 & 1 & 4 & -1 \\ 0 & 1 & 2 & -3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 3 & -2 \\ 0 & -1 & -2 & 3 \\ 0 & 1 & 2 & -3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 3 & -2 \\ 0 & -1 & -2 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

↑ all zeros in row 3

thus, b does not belong to span
 $\times \{v_1, v_2, v_3\}$

$$(ii) \left[\begin{array}{ccc|c} 1 & 1 & 3 & -2 \\ 0 & -1 & -2 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

↑ not a pivot in every column,
 so it is not linearly independent.

3. (20 points) Let A be the matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 3 \\ 0 & 2 & 2 \end{bmatrix}.$$

(i) Describe the set of vectors b in $\mathbb{R}^3$ so that $Ax = b$.

[Your answer should be in the form of equation(s) involving the entries b_1, b_2, b_3 of b .]

(ii) Find a basis for the null space of A .

(i)

$$A = \left[\begin{array}{ccc|c} 1 & 0 & 2 & b_1 \\ 1 & 1 & 3 & b_2 \\ 0 & 2 & 2 & b_3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 2 & b_1 \\ 0 & 1 & 1 & b_2 - b_1 \\ 0 & 2 & 2 & b_3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 2 & b_1 \\ 0 & 1 & 1 & b_2 - b_1 \\ 0 & 0 & 0 & -2(b_2 - b_1) + b_3 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 2 & b_1 \\ 0 & 1 & 1 & b_2 - b_1 \\ 0 & 0 & 0 & b_3 - 2(b_2 - b_1) \end{array} \right]$$

$$x_1 + 2x_3 = b_1$$

$$x_2 + x_3 = b_2 - b_1$$

$$0 = b_3 - 2(b_2 - b_1)$$

(ii) $x_1 = -2x_3$

$$x_2 = -x_3$$

$$\text{nul}(A): \bar{x} = x_3 \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}$$

4. (20 points) Let A be the matrix

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 2 & -3 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

(i) Show that A is invertible and compute its inverse A^{-1} .

(ii) Use A^{-1} to find the solution to $Ax = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$.

(i)

$$\left[\begin{array}{ccc|ccc} 2 & 0 & 0 & 1 & 0 & 0 \\ 2 & -3 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 2 & -3 & 1 & 0 & 1 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 3 & -1 & 1 & -1 & 0 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 & -1 & -3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -1 & 1 & 3 \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$

(ii) $Ax = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

$$x = A^{-1} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$3 \times 3 \times 3 \times 1$
 $= 3 \times 1$

$$x = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/2 + 0 + 0 \\ 0 + 0 + 1 \\ -1 + 0 + 3 \end{bmatrix}$$

$$x = \begin{bmatrix} 1/2 \\ 1 \\ 2 \end{bmatrix}$$

5. (10 points) Suppose T is a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^4$. Suppose

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 8 \end{bmatrix} \quad \text{and} \quad T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix}.$$

Compute $T\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right)$.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 0 \\ 0 & 3 \\ 8 & 0 \end{bmatrix}$$

$$T\begin{bmatrix} 2 \\ 1 \end{bmatrix} = 2\begin{bmatrix} 3 \\ 1 \\ 0 \\ 8 \end{bmatrix} + 1\begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix}$$

$$T\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 0 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix}$$

$$T\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \\ 3 \\ 16 \end{bmatrix} \quad \checkmark$$

6. (10 points) Fill in the blanks.

Suppose T is a linear transformation with standard matrix A .

- (i) T is one-to-one if and only if A has a pivot in every column.
- (ii) T is onto if and only if A has a pivot in every row.